Advancements in Creole NLP: Overcoming Data Scarcity in Creole Syntactic Parsing

Despite their sociolinguistic and historical significance, Creole languages remain underrepresented in computational linguistics, limiting their integration into modern Natural Language Processing (NLP) technologies. Dependency parsers, which analyze sentence structure by identifying grammatical relationships between words, play a critical role in linguistic research and practical applications such as machine translation, speech recognition, and automated text processing. However, the scarcity of annotated corpora, Creole-specific linguistic variation, and structural divergence from their lexifiers pose significant challenges to developing accurate syntactic models for Creole languages. This study advances efforts in Creole NLP by evaluating multiple training paradigms for Haitian Creole and Martinican Creole dependency parsing, comparing baseline monolingual models, concatenated bilingual models, and Multitask Learning (MTL) strategies to assess their impact on generalization, label consistency, and cross-lingual transfer.

We establish baseline models using monolingual training on Haitian Creole[1] and Martinican Creole Universal Dependencies (UD) treebanks, alongside concatenated French-Creole models, which have previously demonstrated strong performance for Martinican Creole parsing (Mompelat et al., 2022). Concurrently, our MTL approach trains on a combination of Haitian Creole and Martinican Creole data splits, allowing models to generalize across varied syntactic constructions. To evaluate model robustness, we compare the results from SuPar's[2] Deep Biaffine parser (Dozat and Manning, 2017) with various transformer-based architectures, including XLM-RoBERTa (Conneau et al. 2019), CamemBERT (Martin et al. 2019), CamemBERTv2 (Antoun et al. 2024) and CreoleEval (Lent et al. 2024). Preliminary results indicate that concatenated training improves parsing performance, particularly for Martinican Creole, while Haitian Creole appears more responsive to RoBERTa-based models. Our results also suggest that CreoleEval models achieve the highest performance on Haitian Creole, surpassing CamemBERT-based approaches. Additionally, concatenated training benefits Martinican Creole more than Haitian Creole, as French-Martinican Creole models consistently outperform monolingual setups. Across training splits, MTL models show greater stability, mitigating performance fluctuations in low-resource settings.

Beyond performance metrics, systematic error analysis reveals that models struggle with low-frequency dependencies and functionally ambiguous structures. The error rate distribution per dependency label[3] also suggests that function words (*case*, *mark*, *det*) are more robustly predicted than content words (*nmod*, *xcomp*, *appos*). A Support vs. F1-score analysis confirms that low-frequency labels, such as *compound:lvc* and *expl*, are rarely predicted correctly, while high-frequency relations, such as *nsubj*, achieve high recall but suffer from interference errors. These misclassifications and annotation inconsistencies reflect both interlanguage interference and cross-

[1] https://github.com/UniversalDependencies/UD_Haitian_Creole-Autogramm/
[2] https://github.com/yzhangcs/parser
[3] See details here https://universaldependencies.org/u/dep/

lingual annotation gaps, highlighting areas where targeted corpus expansion and annotation refinement are needed. These findings therefore open a promising avenue for reinforcement-learning strategies to Creole data enrichment, leveraging Large Language Models for linguistically informed data augmentation. By designing controlled synthetic data generation tasks, we aim to amplify underrepresented syntactic patterns and enhance model robustness.

In conclusion, this research contributes to ongoing advancement in NLP technologies for Creole languages, machine learning for low-resource languages, and AI-driven corpus expansion. By integrating fine-grained augmentation strategies, we aim to bridge the gap between linguistic theory, computational modeling, and practical NLP applications for Creole-speaking communities.

References:

- Antoun, W., Kulumba, F., Touchent, R., de la Clergerie, É., Sagot, B., & Seddah, D. (2024). CamemBERT 2.0: A Smarter French Language Model Aged to Perfection. *arXiv preprint arXiv:2411.08868*.
- Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., ... & Stoyanov, V. (2019). Unsupervised cross-lingual representation learning at scale. *arXiv preprint arXiv:1911.02116*.
- Dozat, T., Qi, P., & Manning, C. D. (2017, August). Stanford's graph-based neural dependency parser at the conll 2017 shared task. In *Proceedings of the CoNLL 2017 shared task: Multilingual parsing from raw text to universal dependencies* (pp. 20-30).
- Lent, H., Tatariya, K., Dabre, R., Chen, Y., Fekete, M., Ploeger, E., ... & Bjerva, J. (2024). CreoleVal: Multilingual multitask benchmarks for creoles. *Transactions of the Association for Computational Linguistics*, *12*, 950-978.
- Martin, L., Muller, B., Suárez, P. J. O., Dupont, Y., Romary, L., de La Clergerie, É. V., ... & Sagot, B. (2019). CamemBERT: a tasty French language model. *arXiv preprint arXiv:1911.03894*.
- Mompelat, L., Dakota, D., & Kübler, S. (2022, October). How to parse a creole: When martinican creole meets French. In *Proceedings of the 29th International Conference on Computational Linguistics* (pp. 4397-4406).
- Sayyed, Z. A., & Dakota, D. (2021, August). Annotations matter: Leveraging multi-task learning to parse UD and SUD. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021* (pp. 3467-3481).